

Лабораторная работа №16

Администрирование локальных сетей

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Информация

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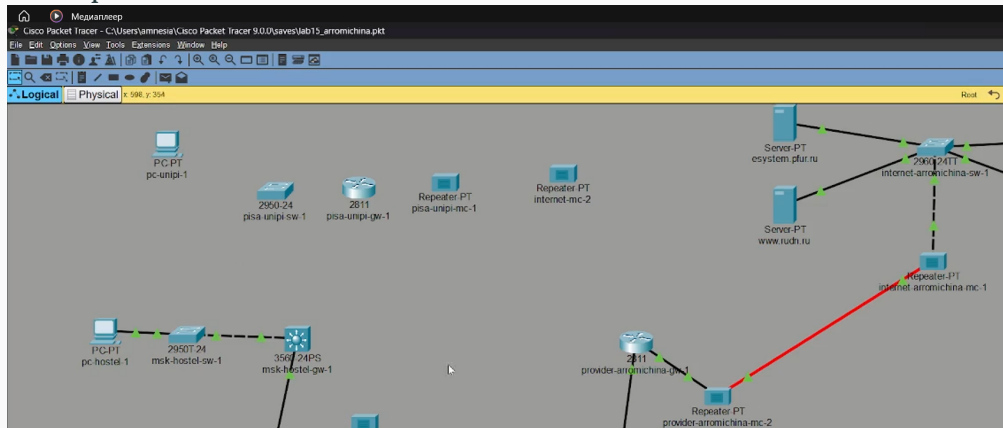
Вводная часть

- Получение навыков настройки VPN-туннеля через незащищённое Интернетсоединение.

Основная часть

Размещение объектов

- Разместим в рабочей области проекта оборудование для сети Университета г. Пиза



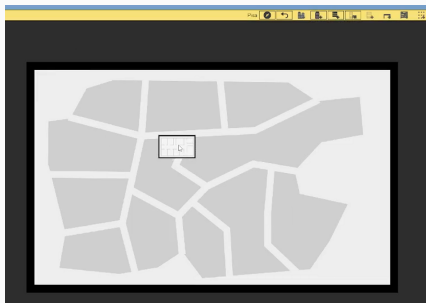
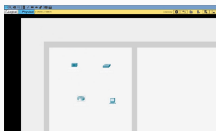
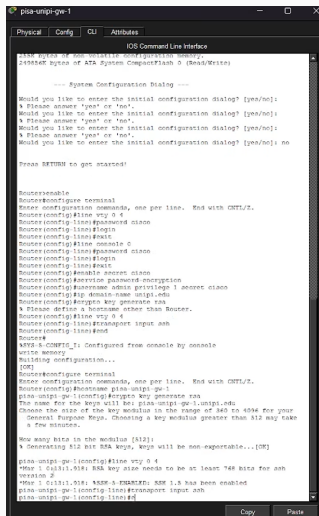


Рис. 1: В физической рабочей области проекта создадим город Пиза, здание Университета г. Пиза

- Переместим туда соответствующее оборудование



Первоначальная настройка



```
IOS Command Line Interface
143K bytes of non-volatile configuration memory.
245856K bytes of ATA System CompactFlash 0 (Read/Write)

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]:
% Please answer 'yes' or 'no'.
Would you like to enter the initial configuration dialog? [yes/no]:
% Please answer 'yes' or 'no'.
Would you like to enter the initial configuration dialog? [yes/no]:
% Please answer 'yes' or 'no'.
Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#line vty 0 4
Router(config-line)#password cisco
Router(config-line)#login
Router(config-line)#exit
Router(config)#line console 0
Router(config-line)#password cisco
Router(config-line)#login
Router(config-line)#exit
Router(config)#enable secret cisco
Router(config)#service password-encryption
Router(config)#username admin privilege 1 secret cisco
Router(config)#ip domain-name unpil.edu
Router(config)#crypto key generate rsa
% Please define a hostname other than Router.
Router(config)#line vty 0 4
Router(config-line)#transport input ssh
Router(config-line)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Write memory
Building configuration...

[OK]
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname pisa-unipi-gw-1
pisa-unipi-gw-1(config)#crypto key generate rsa
The name for the keys will be: pisa-unipi-gw-1.unpil.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

pisa-unipi-gw-1(config)#line vty 0 4
*Mar 1 01:11:01: RSA key also needs to be at least 768 bits for ssh
version 2
*Mar 1 01:11:01: %SSH-5-FRAMEWORK: SSH 1.5 has been enabled
pisa-unipi-gw-1(config-line)#transport input ssh
pisa-unipi-gw-1(config-line)#
```

Рис. 2: Первоначальная настройка маршрутизатора pisa-unipi-gw-1

Первоначальная настройка коммутатора pisa-unip1-sw-1

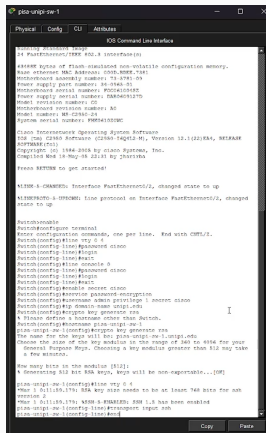
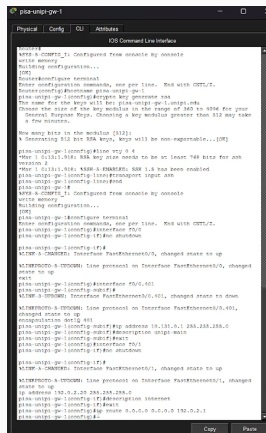


Рис. 3: Первоначальная настройка коммутатора pisa-unipri-sw-1

Настройка интерфейсов маршрутизатора pisa-unipi-gw-1



```
pisa-unipi-gw-1
Physical  Config  GUI  Attributes
IOS Command Line Interface

pisa-unipi-gw-1>enable
pisa-unipi-gw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
pisa-unipi-gw-1(config)#interface fa0/0
The name for the key will be: pisa-unipi-gw-1.unipi-gw-1
Choose the size of the key modulus in the range of 360 to 512 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]:
A Generating 512 bit RSA keys, keys will be non-exportable...[OK]
pisa-unipi-gw-1(config)#ip address 192.0.2.20 255.255.255.0
pisa-unipi-gw-1(config)#no shutdown
pisa-unipi-gw-1(config)#interface fa0/1
The name for the key will be: pisa-unipi-gw-1.unipi-gw-1
Choose the size of the key modulus in the range of 360 to 512 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]:
A Generating 512 bit RSA keys, keys will be non-exportable...[OK]
pisa-unipi-gw-1(config)#ip address 192.0.2.1 255.255.255.0
pisa-unipi-gw-1(config)#no shutdown
pisa-unipi-gw-1(config)#interface fa0/2
The name for the key will be: pisa-unipi-gw-1.unipi-gw-1
Choose the size of the key modulus in the range of 360 to 512 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]:
A Generating 512 bit RSA keys, keys will be non-exportable...[OK]
pisa-unipi-gw-1(config)#ip address 192.0.2.1 255.255.255.0
pisa-unipi-gw-1(config)#no shutdown
pisa-unipi-gw-1(config)#end
```

Рис. 4: Настройка интерфейсов маршрутизатора pisa-unipi-gw-1

Настройка интерфейсов коммутатора pisa-unip1-sw-1

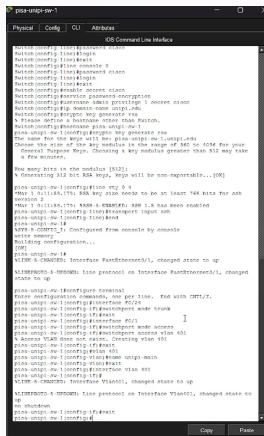
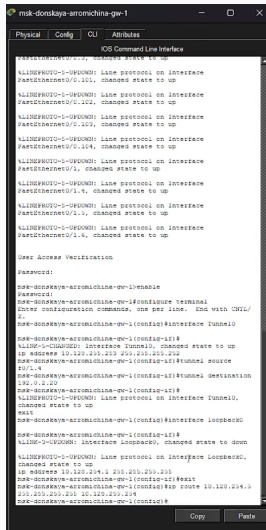


Рис. 5: Настройка интерфейсов коммутатора pisa-unip1-sw-1

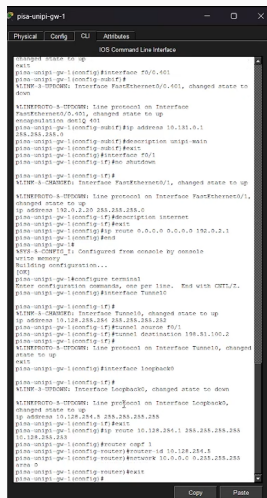
Настройка маршрутизатора msk-donskaya-gw-1



```
msk-donskaya-aromichina-gw-1
Physical Config CLI Attributes
IOS Command Line Interface
FastEthernet0/0.1: changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/0.101, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/0.102, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/0.103, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/0.104, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/1.4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/1.5, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/1.6, changed state to up
User Access Verification
Password:
msk-donskaya-aromichina-gw-1#enable
Password:
msk-donskaya-aromichina-gw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/
Z.
msk-donskaya-aromichina-gw-1(config)#interface Tunnel0
msk-donskaya-aromichina-gw-1(config-if)#
%LINE-5-CHANGED: Interface Tunnel0, changed state to up
ip address 10.128.255.255 255.255.255.255
msk-donskaya-aromichina-gw-1(config-if)#tunnel source
10/1.4
msk-donskaya-aromichina-gw-1(config-if)#tunnel destination
10.128.255.255
msk-donskaya-aromichina-gw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel0,
changed state to up
exit
msk-donskaya-aromichina-gw-1(config)#interface loopback0
msk-donskaya-aromichina-gw-1(config-if)#
%LINE-5-UPDOWN: Interface Loopback0, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0,
changed state to up
ip address 10.128.255.255 255.255.255.255
msk-donskaya-aromichina-gw-1(config-if)#exit
msk-donskaya-aromichina-gw-1(config)#ip route 10.128.255.0
255.255.255.255 10.128.255.255
msk-donskaya-aromichina-gw-1(config)#
```

Рис. 6: Настройка маршрутизатора msk-donskaya-gw-1

Настройка интерфейсов маршрутизатора pisa-unipi-gw-1



```
pisa-unipi-gw-1
Physical  Config  CLI  Attributes
IOS Command Line Interface

pisa-unipi-gw-1>
pisa-unipi-gw-1>
pisa-unipi-gw-1(config)#interface fa0/0/401
pisa-unipi-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.401, changed state to
down
%LINKPROTO-5-IFDOWN: Line protocol on Interface
FastEthernet0/0.401, changed state to up
%configuration done! 401
pisa-unipi-gw-1(config-subif)#ip address 10.128.0.1
255.255.255.0
pisa-unipi-gw-1(config-subif)#description unipi-main
pisa-unipi-gw-1(config-subif)#exit
pisa-unipi-gw-1(config)#interface fa0/1
pisa-unipi-gw-1(config-if)#no shutdown

pisa-unipi-gw-1(config-if)#
%LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
%LINKPROTO-5-IFDOWN: Line protocol on Interface FastEthernet0/1,
changed state to up
ip address 192.0.2.20 255.255.255.0
pisa-unipi-gw-1(config-if)#description internet
pisa-unipi-gw-1(config-if)#exit
pisa-unipi-gw-1(config)#ip route 0.0.0.0 0.0.0.0 192.0.2.1
pisa-unipi-gw-1(config)#end
pisa-unipi-gw-1#
%SYS-5-CONFIG_1: Configured from console by console
write memory
Building configuration...
[OK]
pisa-unipi-gw-1(config)#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
pisa-unipi-gw-1(config)#interface Tunnel0

pisa-unipi-gw-1(config-if)#
%LINK-3-UPDOWN: Interface Tunnel0, changed state to up
pisa-unipi-gw-1(config-if)#tunnel source fa0/1
pisa-unipi-gw-1(config-if)#tunnel destination 198.51.100.2
pisa-unipi-gw-1(config-if)#
%LINKPROTO-5-IFDOWN: Line protocol on Interface Tunnel0, changed
state to up
pisa-unipi-gw-1(config)#interface loopback0

pisa-unipi-gw-1(config-if)#
%LINK-3-UPDOWN: Interface loopback0, changed state to down
%LINKPROTO-5-IFDOWN: Line protocol on Interface loopback0,
changed state to up
ip address 10.128.254.1 255.255.255.255
pisa-unipi-gw-1(config-if)#exit
pisa-unipi-gw-1(config)#ip route 10.128.254.1 255.255.255.255
10.128.255.255
pisa-unipi-gw-1(config)#router ospf 1
pisa-unipi-gw-1(config-router)#router-id 10.128.254.1
pisa-unipi-gw-1(config-router)#network 10.0.0.0 0.255.255.255
area 0
pisa-unipi-gw-1(config-router)#exit
pisa-unipi-gw-1(config)#
```

Рис. 7: Настройка интерфейсов маршрутизатора pisa-unipi-gw-1

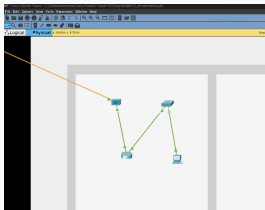


Рис. 8: Проверка

Вывод

- Мы смогли получить навыки настройки VPN-туннеля через незащищённое Интернетсоединение.